

Imperial College London
Department of Earth Science and Engineering
MSc in Applied Computational Science and Engineering

Independent Research Project
Project Plan

Parameter Sensitivity Analysis and Machine Learning Prediction of Tidal Disruption Outcomes in SPH Simulations of Martian Moon Formation

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Abstract

This project investigates how orbital, physical, and numerical parameters influence tidal-disruption outcomes in SPH simulations of asteroid encounters with Mars. Tidal disruption is a proposed mechanism for generating debris that may contribute to the formation of Phobos and Deimos, but the relationships between encounter conditions and debris structures remain incompletely characterized. The project will develop a computational pipeline to process existing SPH/HDF5 simulation outputs, extract fragment properties, and connect Friends-of-Friends group information with physical snapshot data. Key derived outcomes will include fragment distributions, bound debris mass fraction, bound fragment count, and largest bound fragment mass. Exploratory analysis and parameter-sensitivity methods will identify dominant controls on disruption outcomes. Machine-learning modelling will begin with baseline predictors to establish whether outcomes contain learnable structure, followed by validation, model comparison, and optimisation where appropriate. The expected deliverables are a validated structured dataset, quantitative sensitivity results, predictive-model benchmarks, and a physical interpretation of parameter regimes relevant to debris retention around Mars.

1. Introduction

1.1. Problem Description and Research Questions

Mars's moons, Phobos and Deimos, have an uncertain origin. Their asteroid-like appearance suggests capture, but their low-inclination, near-circular orbits are difficult to explain by simple capture alone. Impact-generated debris disks provide another explanation, but require specific impact conditions and subsequent disk evolution. A recent alternative is disruptive partial capture, in which an asteroid passing close to Mars is tidally disrupted and some fragments are retained in Mars orbit (Kegerreis et al., 2024).

This IRP asks: how do orbital and physical parameters influence tidal-disruption outcomes; which parameters most strongly control fragment formation and bound debris mass; and can machine-learning models predict disruption outcomes across the sampled parameter space? The computational challenge is to extract, organise, and model an existing SPH simulation campaign, rather than to run new simulations from scratch.

1.2. Significance

This work matters because it connects Martian moon formation with a practical computational-science problem. If tidal disruption can produce retained debris, then identifying the encounter conditions that favour bound material helps evaluate whether this pathway is plausible. More broadly, the project tests whether raw particle simulation outputs can be transformed into reliable structured datasets for sensitivity analysis and predictive modelling.

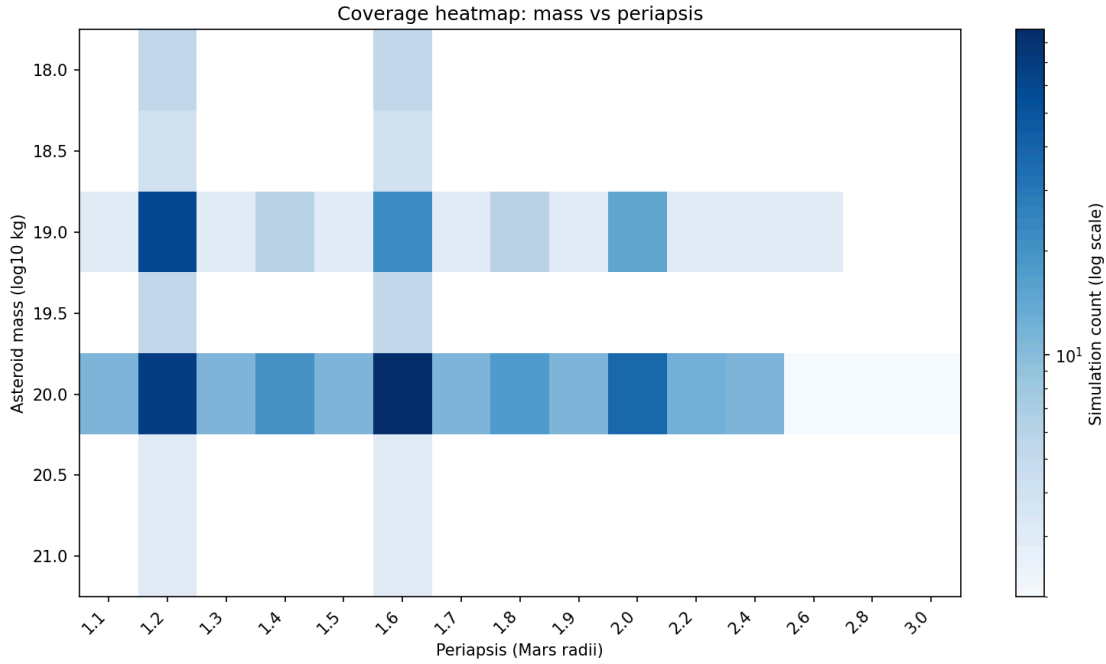


Figure 1. Existing simulation coverage across asteroid mass and periapsis. Uneven sampling motivates both full-dataset analysis and controlled subsets.

1.3. Literature Review

Several formation mechanisms have been proposed for Phobos and Deimos. Giant-impact studies show that impacts on Mars can create debris disks massive enough to form small moons (Citron et al., 2015; Canup & Salmon, 2018), while Rosenblatt et al. (2016) argued that transient inner moons could stir an outer disk and support accretion of small outer moons. Other work proposes that the moons may be remnants of a disrupted common progenitor (Bagheri et al., 2021). These studies demonstrate that the origin problem is active, but they also show that debris production, capture, and later orbital evolution remain difficult to connect.

Kegerreis et al. (2024) provide the most direct motivation for this project by modelling disruptive partial capture of an asteroid near Mars. Their work suggests that tidal disruption can capture a non-negligible fraction of asteroid material over a range of periapsis distances, velocities, masses, spins, and orientations. This IRP builds on that scenario by analysing the SPH disruption outputs themselves: instead of asking only whether capture is possible, it asks which input parameters control fragment and bound-debris outcomes.

SPH is appropriate because it is a mesh-free Lagrangian method widely used for astrophysical, planetary, and impact-like simulations, although its numerical choices and post-processing steps require careful interpretation (Price, 2012). However, fragment identification introduces post-processing uncertainty. FoF algorithms group particles using a linking length, and previous FoF studies show that inferred structures can depend strongly on this parameter (More et al., 2011). Therefore, FoF fragment counts are useful proxies, but they should not be treated as final physical moon-forming outcomes.

Machine learning is increasingly used to extract patterns from simulation outputs and identify important controls in complex physical systems (Brunton et al., 2020). In this project, ML is not intended to replace physical analysis. It is used diagnostically: to test predictability, compare targets, inspect feature importance, and expose where results are limited by uneven sampling or post-processing choices.

1.4 Objectives

Table 1. Objectives and measurable success criteria

Objective	Success criterion
Build a fragment and bound-debris extraction pipeline	Generate structured tabular datasets for initial conditions and FoF-derived outcomes, including bound/unbound metrics for the full dataset
Quantify parameter sensitivity	Identify the dominant physical and numerical controls using EDA on both full and controlled subsets
Develop and validate predictive models	Train baseline and selected tuned models for FoF-derived and bound-aware targets, with validation, residual analysis, feature-importance diagnostics, and reliability checks
Distinguish physics from post-processing	Compare model behaviour with and without FoF linking length
Deliver a defensible interpretation	Produce a final report explaining results, limitations, and implications for the disruptive-capture scenario

2. Methodology

2.1 Data and Simulation Outputs

The project uses existing SPH/HDF5 simulations of asteroid encounters with Mars. FoF outputs provide group identities and fragment-size proxies, while physical snapshots provide particle coordinates, velocities, and masses. These sources will be joined into simulation-level and fragment-level tables.

2.2 Fragment and Bound-Debris Extraction

The extractor will compute fragment statistics and classify material as bound or unbound relative to Mars using specific orbital energy. The main targets will be bound mass fraction, bound fragment count, and largest bound fragment mass. These are closer to the formation question than FoF counts alone, although bound material is only a necessary, not sufficient, condition for a long-term moon-forming disk.

2.3 Parameter Sensitivity Analysis

Exploratory analysis will compare the full dataset with controlled subsets that fix timestep, dominant resolution, and FoF linking length. This is needed because the data is unevenly sampled and because numerical or post-processing choices can obscure physical trends.

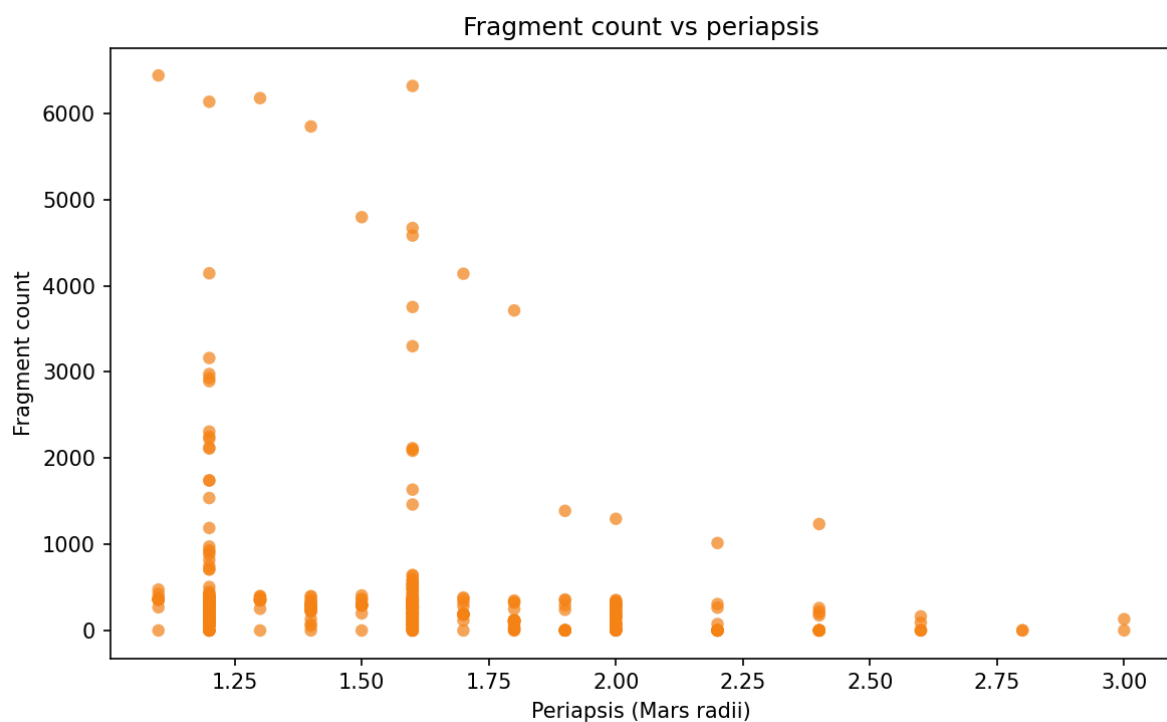


Figure 2. Current FoF trend between periapsis and fragment count. Closer encounters tend to produce more detected fragments.

2.4 Machine Learning Prediction and Validation

ML models will use one row per simulation. Baseline models will establish reference performance, followed by model comparison and tuning where targets show learnable structure. Validation will use train-test performance, residual analysis, feature importance, and checks with/without FoF linking length.

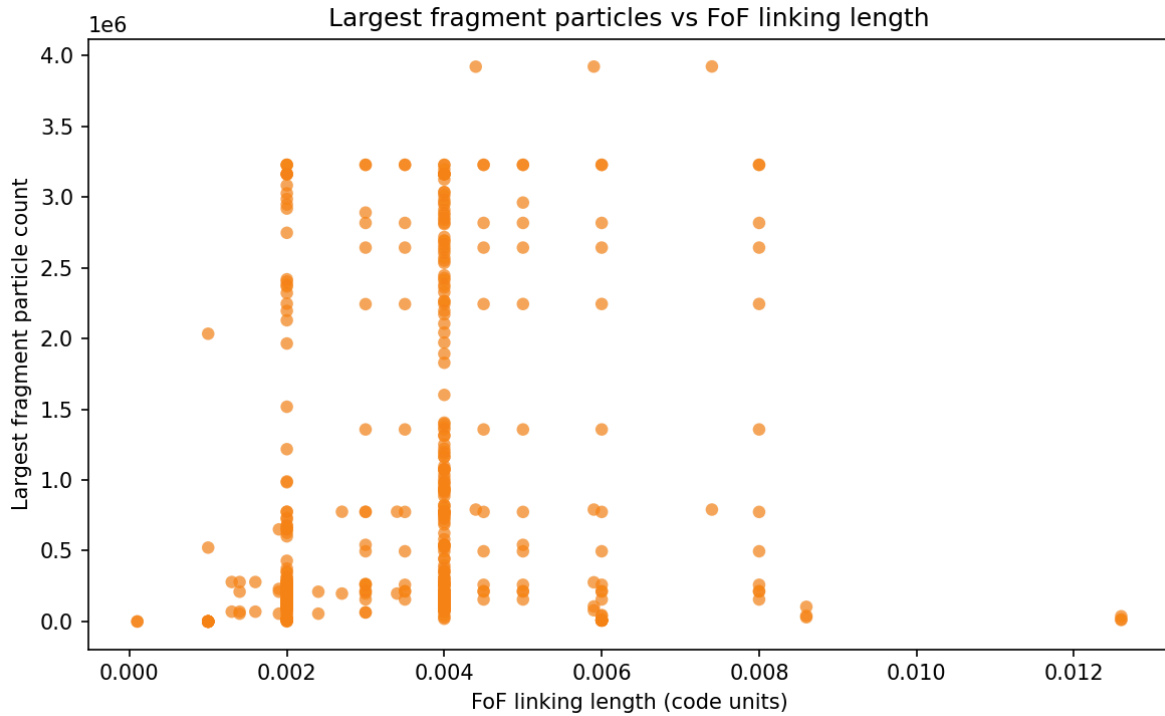


Figure 3. Largest detected fragment particle count as a function of FoF linking length. The scatter suggests that FoF linking length may affect fragment measurements, but it is not the only controlling factor. This motivates checking post-processing effects by comparing model results with and without FoF linking length.

2.5 Limitations and Alternative Approaches

Running a new simulation campaign would give cleaner coverage, but would exceed the three-month IRP scope. A purely descriptive project would be safer but would miss the chance to test predictability. The chosen approach balances feasibility, physical interpretation, and computational analysis.

3. Progress to Date

Preliminary work has produced FoF-based extraction for 489 simulations, initial EDA, and baseline ML diagnostics. Early results suggest that periapsis and mass influence disruption outcomes, while FoF linking length may affect fragment measurements on high linking length. These results support feasibility but also motivate the shift from FoF outcomes to bound-aware physical metrics.

4. Future Plan, Feasibility, and Risks

Table 2. Planned timeline for the remaining IRP period

Period	Main tasks	Milestones
June 2026	Extract bound-aware outcomes on HPC, validate schemas and completeness	Full bound-aware CSV tables available
July 2026	Run updated EDA and ML, compare FoF-only and bound-aware targets	Sensitivity and model benchmarks complete
August 2026	Interpret physical implications, refine figures, write final report	Final narrative, limitations, and conclusions complete

Main risks are manageable. HPC or data-access delays will be mitigated by keeping local analysis scripts ready and transferring compact CSV outputs. Sparse or noisy bound-aware targets will be handled by analysing both full and controlled subsets. If ML adds limited value, it will remain a diagnostic tool rather than the sole contribution. If FoF choices dominate, models will be compared with and without linking length, and physical interpretation will prioritise bound-aware targets.

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